

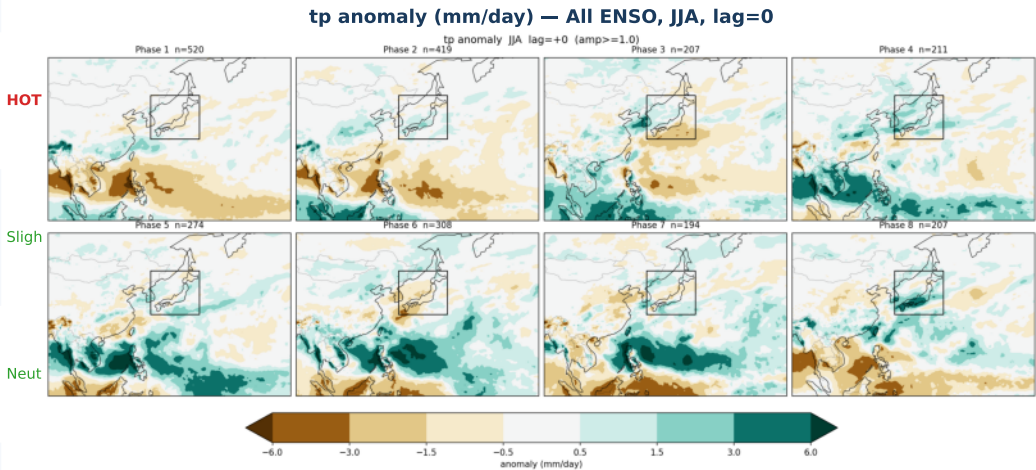
Japan Summer MJO Cheat Sheet

ERA5 JJA | t2m 1994–2023 / tp 1979–2023 | RMM amp ≥ 1.0 | lag = 0

ALL ENSO

Box: Japan 30–46 N, 128–146 E | ** p<0.05 * p<0.10 (two-sided t-test, n eff = n/5)
t2m sig: Japan box-mean (σ=1.4°C) tp sig: best same-sign sub-region

Ph	Convection	t2m (°C)	tp (mm/d)	Japan tendency
1	W. Hemisphere / Africa	-0.14	-0.16 *	Slightly cool / E. Japan dry
2	Western Indian Ocean	-0.07	+0.21	Near-neutral / Slightly wet
3	Eastern Indian Ocean	+0.46 **	-0.55	
4	Maritime Continent (W)	+0.12	+0.61	
5	MC / Philippine Sea	+0.04	+0.32	
6	Western Pacific	+0.14	-0.61 **	Slightly warm / W. Japan DRY ★
7	Central Pacific	-0.21	+0.24	Cool (N. Japan) / Slightly wet
8	Eastern Pacific / W. Hem.	-0.07	+1.06 **	Slightly cool / WETTEST ★

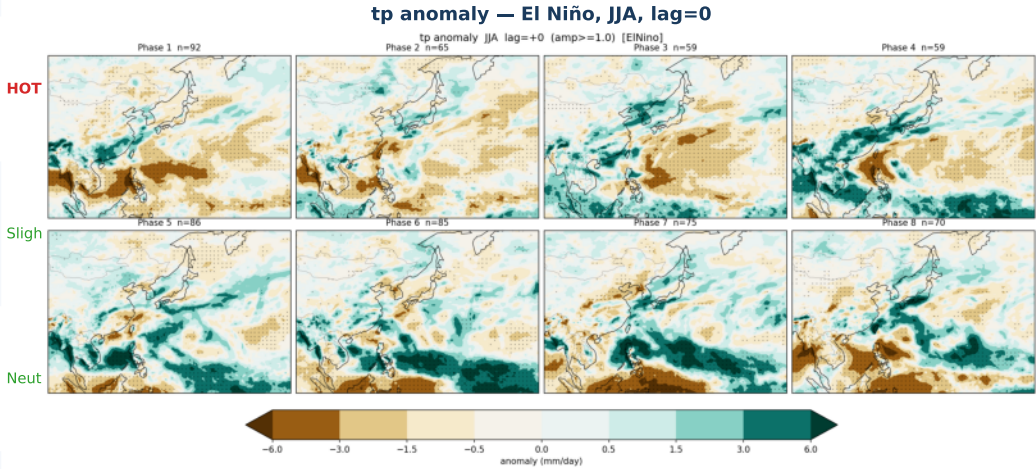


[!] n = 28-89 per phase (El Nino years); no cells formally significant

Japan Summer MJO Cheat Sheet

EL NIÑO REGIME

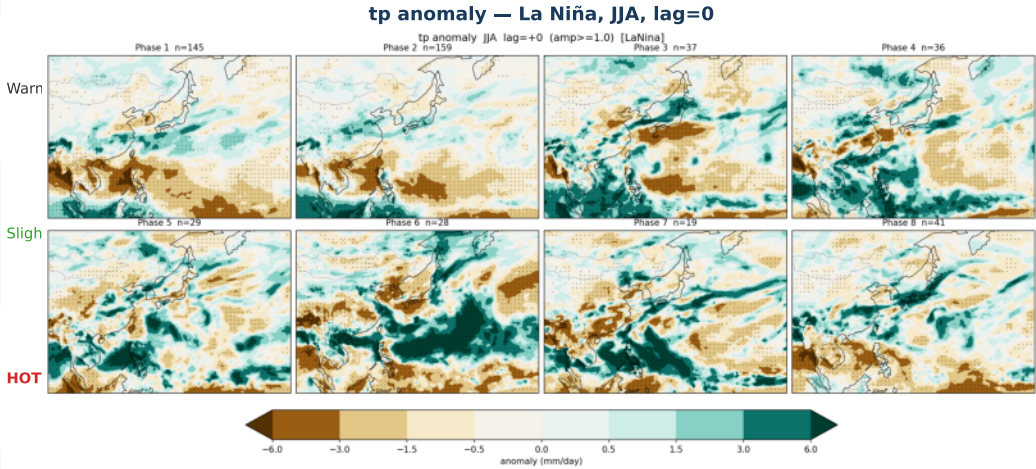
Ph	Convection	t2m (°C)	tp (mm/d)	Japan tendency
1	W. Hemisphere / Africa	+0.10	-0.60	Near-neutral / Dry
2	Western Indian Ocean	+0.46	+0.31	Warm / Wet ★
3	Eastern Indian Ocean	+0.86	+0.18	
4	Maritime Continent (W)	+0.21	+0.85	
5	MC / Philippine Sea	-0.01	+0.53	
6	Western Pacific	+0.20	-0.34	Slightly warm / Dry (dampened)
7	Central Pacific	-0.17	+0.57	Slightly cool / Slightly wet
8	Eastern Pacific / W. Hem.	-0.07	+1.11	Near-neutral / WETTEST ★



Japan Summer MJO Cheat Sheet

LA NIÑA REGIME

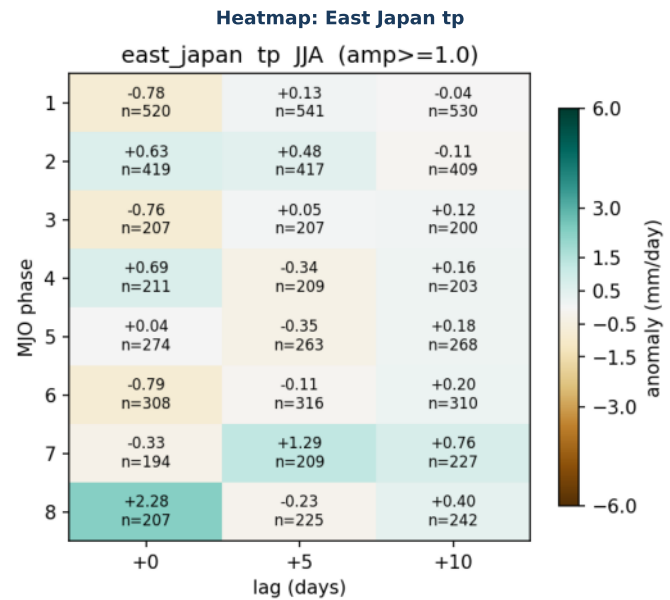
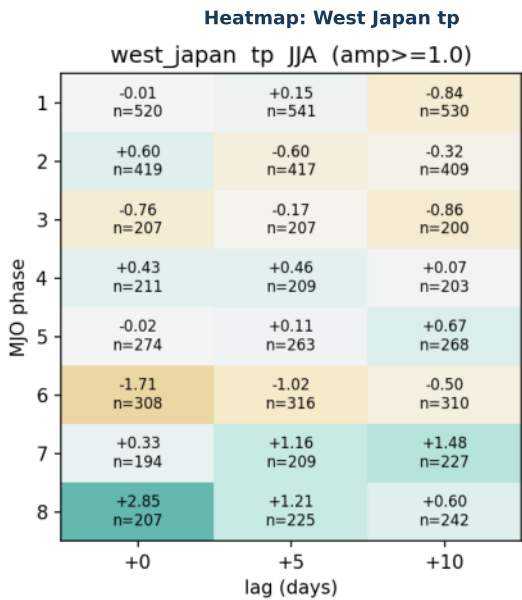
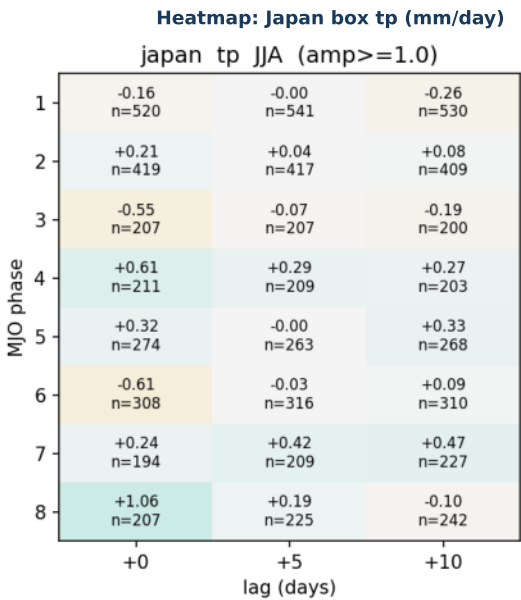
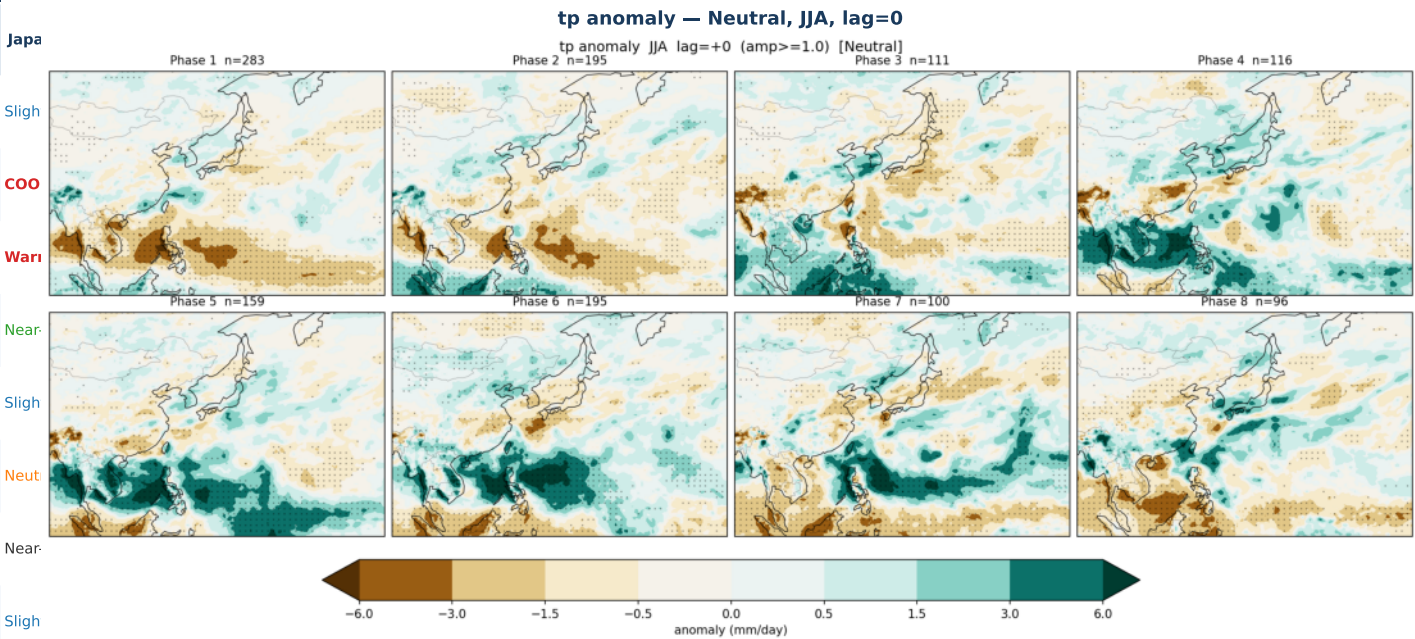
Ph	Convection	t2m (°C)	tp (mm/d)	Japan tendency
1	W. Hemisphere / Africa	-0.23	+0.36	Slightly cool / Slightly wet
2	Western Indian Ocean	+0.20	+0.25	Slightly warm / Slightly wet
3	Eastern Indian Ocean	+0.48	-0.18	
4	Maritime Continent (W)	+0.16	+0.34	
5	MC / Philippine Sea	+0.71	-0.14	
6	Western Pacific	+0.47	-0.80 **	Warm / DRIEST ★
7	Central Pacific	-0.75	+0.68	COOLEST (fragile[!]) / Wet
8	Eastern Pacific / W. Hem.	+0.42	+1.75	Slightly warm / WETTEST ★



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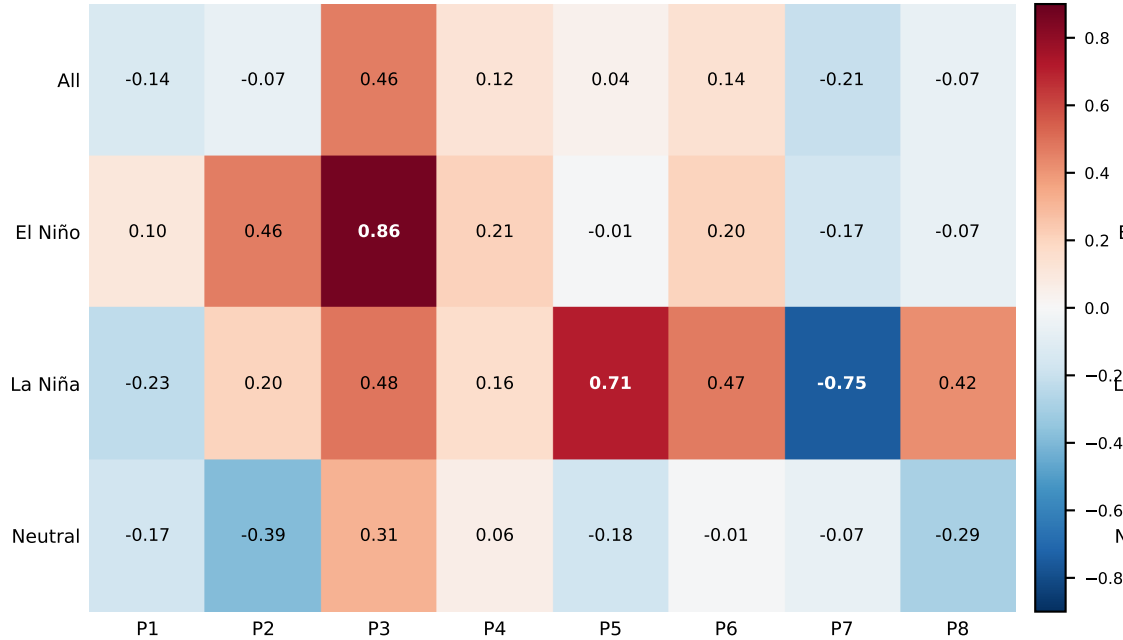
NEUTRAL REGIME

Ph	Convection	t2m (°C)	tp (mm/d)	
1	W. Hemisphere / Africa	-0.17	-0.29 **	Sligh
2	Western Indian Ocean	-0.39 **	+0.15	COO
3	Eastern Indian Ocean	+0.31	-1.06 **	Wari
4	Maritime Continent (W)	+0.06	+0.57	Near
5	MC / Philippine Sea	-0.18	+0.29	Sligh
6	Western Pacific	-0.01	-0.70 **	Neut
7	Central Pacific	-0.07	-0.10	Near
8	Eastern Pacific / W. Hem.	-0.29	+0.72	Sligh

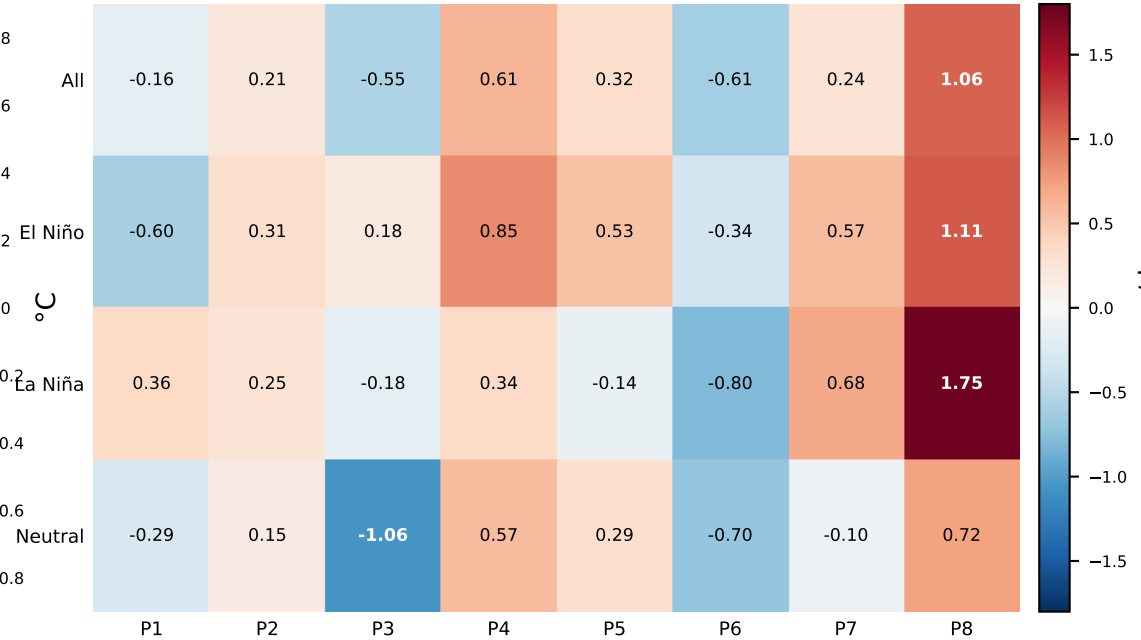


Japan Summer MJO Cheat Sheet

t2m anomaly (°C) — Japan box, JJA, lag=0



tp anomaly (mm/day) — Japan box, JJA, lag=0



Key findings:

P3 (E. Indian Ocean): Hottest + Dry
robust in ALL ENSO (t2m +0.46C **)
P8 (E. Pacific): Strongest wet
ALL ENSO (tp +1.06** ; La Nina +1.75)
P6 (W. Pacific): Driest - all ENSO
W. Japan -1.71 **
P7 (Central Pac.): Coolest N. Japan
El Nino: amplifies P3 heat (+0.86C),
P4 wet (+0.85 mm/d)
La Nina: hot shifts to P5-6;
P8 precip peaks (+1.75 mm/d)
Neutral: P3 dry Neutral-only (-1.06 **)
P2 cool (-0.39C **)

Significance notes:

** p<0.05 * p<0.10 (two-sided t-test)
t2m: Japan box-mean, sigma_box=1.4C, n_eff=n/5
tp: best same-sign sub-region (W./E./N. Japan
or Okinawa), sigma_box=gridpt sigma/sqrt(6)
El Nino/La Nina layers (n=28-89): no cells
formally significant
[!] La Nina Phase 7: n=19, directional only